Fiscal Year:	FY 2013	**Task Last Updated:**	FY 01/22/2013
PI Name:	Kronenberg, Amy Sc.D.		
Project Title:	Comparative Analysis of Charged Particle-Induced Autosomal Mutagenesis in Murine Tissue and Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-8099	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Radiobiology NNJ11ZSA001N
Start Date:	03/14/2012	**End Date:**	03/31/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	4
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Turker, Mitchell (Oregon Health & Science University) Grygoryev, Dmytro (Oregon Health & Science University) Grossi, Gianfranco (Universita Federico II) Dan, Cristian (Oregon Health & Science University) Gauny, Stacey (Lawrence Berkeley National Laboratory) Eckelmann, Bradley (Oregon Health & Science University)		
Grant/Contract No.:	NNJ12HB88I		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Astronauts in space are exposed to a variety of charged particle radiations, some of which travel at very high velocities that can penetrate spacecraft, damage tissues and increase the risk of cancer. This proposal addresses two cancer development processes, genomic instability and aneuploidy, links them to each other and to specific mechanisms of mutagenesis at an endogenous autosomal locus in epithelial cells. The aims proposed use a comparative biology approach and a transgenic mouse model that involves both in vitro and in situ approaches. This strategy enables quantitative, coupled analyses of early changes in cells that survive the exposure and a direct test of the relevance of these changes to the development of persistent changes in situ. The studies address the radiation quality dependence of genomic instability and aneuploidy focusing on charged particle beams spanning a critical LET range from 25 keV/µm to 151 keV/µm, and will define the shape of the dose-response for these endpoints, particularly at low fluence. The rationale for working in this range of LET is the observation of increased biological effectiveness for cancer relevant endpoints in our previous studies, including mutagenesis in human and rodent cells, and work by others concerning Harderian gland tumorigenesis in mice. The investigations test the hypothesis that the induction of genomic instability and aneuploidy will increase as a function of ionization density for low fluence exposures, leading to genome-wide alterations in viable, mutant cells found in normal tissue. This project will define links between the processes that lead to early establishment of genomic instability, aneuploidy, and mechanisms of autosomal mutagenesis found in many kinds of human epithelial cancers. These quantitative, mechanism-based studies will aid in the extrapolation of results across species to help reduce the uncertainties in cancer risk projections for astronauts exposed to spaceflight radiations.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	This research examines the effects of charged particle radiations with different ionization densities using normal mammalian epithelium. The research is designed to help determine links between cancer development processes at relatively low doses. This work is of relevance to naturally occurring Earth-based exposures to densely ionizing radiations such as radon. It will also inform our understanding of the mechanisms underlying potential risks to normal tissues associated with the implementation of charged particle radiation therapy.		
Task Progress:	This project was initiated in Spring 2012. During the first year of funding, we conducted replicate experiments to determine the dose-response for autosomal mutagenesis following exposure to graded doses of 240 MeV/amu Si ions (LET~80 keV/µm) using cultured kidney cells. A dose-dependent increase in Aprt mutations was noted for B6D2F1 Aprt+/- cells exposed in Spring 2012 and in Fall 2012 at the NSRL accelerator. Experiments were also initiated in Summer 2012 with B6D2F1 Aprt+/- mice exposed to graded doses of 263 MeV/amu Si ions designed to mimic the in vitro studies while adjusting for the geometry of the exposures. Also in Fall 2012, dose-response studies were completed for Aprt+/- kidney epithelial cells exposed to 250 MeV/amu O ions (LET~25 keV/µm). Dose-dependent cell killing was observed for these oxygen ions in three replicate experiments. These results will be compared and contrasted with similar data obtained with different charged particle beams to determine the relative toxicity of different ion species to kidney epithelium. In addition to the quantitative studies described above, a large series of independent cultures of B6D2F1 Aprt+/- kidney cells were exposed to very low doses of 1 GeV/amu Fe ions (Spring 2012) and these will supplement ongoing analyses of mutational signatures detected for higher dose exposures (maximum of 1 Gy). The goal is to identify signatures of exposure that may support our central hypothesis that links together several cancer development processes including centrosome abnormalities, aneuploidy, and molecular mechanisms of mutagenesis. Indications to date suggest a link between certain types of mutational events on mouse chromosome 8 and additional genome-wide changes. Large collections of clonal isolates of non-mutant cells and of Aprt-deficient mutants were archived at intermediate and high Fe ion dose levels to evaluate the all endpoints listed and additional independent cultures were archived following exposure to the lowest Fe ion dose proposed. These cultures are source material for clonal isolates that will be developed as needed. The characterization of clones from 1 GeV/amu Fe ion irradiated B6D2F1 Aprt+/- kidneys exposed previously to varying doses (maximum of 1 Gy) is also underway using genome-wide molecular approaches as well as fluorescence in situ hybridization approaches.		

	During Fall 2012, a large series of replicate, independent cultures of B6D2F1 Aprt+/- kidney cells was irradiated with the anchor dose of 240 MeV/amu Si ions to collect both non-mutant and Aprt-deficient mutant clones for further characterization of the various cancer development processes described above. These clones are being expanded and archived, with 58 Aprt mutants and 20 non-mutants collected at 1.6 Gy. This work is complemented by the follow-on harvest and characterization of non-mutant and Aprt-deficient mutant clones from the kidneys of B6D2F1 Aprt+/- mice exposed in Summer 2012 at the NSRL accelerator.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Turker M, Kronenberg A, Grygoryev D, Dan C, Ohrlich A, Eckelmann B, Gauny S. "Toxicity and mutagenesis in three mouse tissues exposed to accelerated Fe ions, Ti ions, or protons." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012
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Abstracts for Journals and Proceedings	Snijders AM, Mannion BJ, Rydberg B, Leung SG, Kronenberg A, Wiese C. "An in-vivo-like multi-cellular skin model system for the analysis of radiation quality effects." 58th Annual Meeting of the Radiation Research Society, San Juan, Puerto Rico, September 30-October 3, 2012. 58th Annual Meeting of the Radiation Research Society, San Juan, Puerto Rico, September 30-October 3, 2012. Presentation number PS5-16. Abstract available at: http://www.abstractsonline.com/Plan/ViewAbstract.aspx?sKey=f8c8e816-7bf5-4085-94f8-f076ca29978a&cKey=44d741e3-5d6b-4095-b05b-8e715d568b75&mKey={1514526B-EDA6-42BC-948B-C58023E5A94E} ; accessed 1/24/13. , Oct-2012
Articles in Peer-reviewed Journals	Kronenberg A, Cucinotta FA. "Space radiation protection issues." Health Physics. 2012 Nov;103(5):556-67. http://dx.doi.org/10.1097/HP.0b013e3182690caf ; PubMed PMID: 23032885 , Nov-2012
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